Game Score Prediction Summary Notebook

This notebook turns the Streamlit workflow in this repository into a single reproducible analysis. It covers:

- exploratory data analysis (EDA)
- model training for linear, ridge, and lasso regression
- model testing on a held-out split
- a supplementary Gaussian Naive Bayes classification experiment
- metric explanations and model visualizations

The notebook uses the merged GRIVG dataset in `datasets/merged_grivg_data.csv` and keeps the same cleaning logic used by the app wherever possible.

In [1]:
```
from pathlib import Path
import re

from IPython.display import display
import numpy as np
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LinearRegression, LassoCV, RidgeCV
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    f1_score,
    mean_absolute_error,
    mean_squared_error,
    precision_score,
    r2_score,
    recall_score,
)
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

pd.set_option("display.max_columns", 100)
pd.set_option("display.float_format", lambda value: f"{value:.3f}")

RANDOM_STATE = 45
TEST_SIZE = 0.30
NB_HIGH_SCORE_THRESHOLD = 9.0
IGN_HIGH_THRESHOLD = 0.95
IGN_LOW_THRESHOLD = 0.9
DATASET_PATH = Path("../datasets/merged_grivg_data.csv") if Path("../datasets/me
```

In [2]:
```
def clean_column_name(column):
    column = str(column).strip()
    column = re.sub(r"[\s\-]+", "_", column)
    column = re.sub(r"[^0-9a-zA-Z_]", "", column)
    column = re.sub(r"_+", "_", column)
    return column.strip("_")


def parse_percentage(series):
    cleaned = (
        series.astype(str)
        .str.replace("%", "", regex=False)
        .str.strip()
        .replace({"": np.nan, "nan": np.nan, "None": np.nan, "Unknown": np.nan})
    )
    values = pd.to_numeric(cleaned, errors="coerce")
    if not values.dropna().empty and values.dropna().max() > 1:
        values = values / 100
    return values


def preprocess_merged_dataset(df):
    cleaned = df.copy()
    cleaned.columns = [clean_column_name(column) for column in cleaned.columns]
    cleaned = cleaned.dropna(axis=1, how="all")
    cleaned = cleaned.loc[:, ~cleaned.columns.str.startswith("Unnamed")]

    for column in cleaned.select_dtypes(include="object").columns:
        cleaned[column] = cleaned[column].astype(str).str.strip()
        cleaned[column] = cleaned[column].replace(
            {"": np.nan, "nan": np.nan, "None": np.nan, "Unknown": np.nan}
        )

    numeric_columns = [
        "Playable",
        "Sexualization",
        "Protagonist",
        "Protagonist_Non_Male",
        "Relevant_males",
        "Relevant_no_males",
        "Total_team",
        "female_team",
        "Metacritic",
        "Destructoid",
        "IGN",
        "GameSpot",
        "Avg_Reviews",
        "Sexualized_clothing",
        "Trophy",
        "Damsel_in_Distress",
        "Sexualized_Cutscenes",
        "Total",
        "PEGI",
    ]
    for column in numeric_columns:
        if column in cleaned.columns:
            cleaned[column] = pd.to_numeric(cleaned[column], errors="coerce")

    if "Percentage_non_male" in cleaned.columns:
```

```
        cleaned["Percentage_non_male_num"] = parse_percentage(cleaned["Percentag
    if "Team_percentage" in cleaned.columns:
        cleaned["Team_percentage_num"] = parse_percentage(cleaned["Team_percenta

    if "Customizable_main" in cleaned.columns:
        cleaned["Customizable_main_flag"] = cleaned["Customizable_main"].map(
            {"Yes": 1, "No": 0, "Non_Binary": 1, "Non-Binary": 1}
        )

    if "Romantic_Interest" in cleaned.columns:
        cleaned["Romantic_Interest_flag"] = cleaned["Romantic_Interest"].map(
            {"Yes": 1, "Opt": 1, "No": 0}
        )

    if "Age" in cleaned.columns:
        cleaned["Age_numeric"] = pd.to_numeric(cleaned["Age"], errors="coerce")

    if "Age_range" in cleaned.columns:
        age_label_map = {
            "Infant": 3,
            "Child": 10,
            "Teenager": 16,
            "Young_adult": 21,
            "Young_Adult": 21,
            "Young adult": 21,
            "Adult": 30,
            "Middle_aged": 50,
            "Middle_Aged": 50,
            "Middle-aged": 50,
            "Elderly": 70,
        }
        if "Age_numeric" not in cleaned.columns:
            cleaned["Age_numeric"] = np.nan
        cleaned["Age_numeric"] = cleaned["Age_numeric"].fillna(
            cleaned["Age_range"].map(age_label_map)
        )

    if "Release" in cleaned.columns:
        release_dt = pd.to_datetime(cleaned["Release"], format="%b-%y", errors="
        cleaned["Release_Date"] = release_dt
        cleaned["Release_Year"] = release_dt.dt.year
        cleaned["Release_Month"] = release_dt.dt.month

    return cleaned
```

In [3]:

```
raw_df = pd.read_csv(DATASET_PATH)
df = preprocess_merged_dataset(raw_df)

print(f"Dataset path: {DATASET_PATH}")
print(f"Rows: {df.shape[0]}")
print(f"Columns: {df.shape[1]}")

missing_summary = (
    pd.DataFrame({
        "dtype": df.dtypes.astype(str),
        "missing_count": df.isna().sum(),
        "missing_rate": (df.isna().mean() * 100).round(2),
    })
    .sort_values(["missing_count", "dtype"], ascending=[False, True])
)
```

```
display(df.head())
display(missing_summary.head(15))
```

```
Dataset path: ..\datasets\merged_grivg_data.csv
Rows: 637
Columns: 52
```

	Name	Gender	Game	Age	Age_range	Playable	Sexualization	
0	Farah	Female	CODMW	27	Adult	1	0	CO
1	Protagonist	Custom	PSS	Teenager	Teenager	1	0	PSS_
2	Magnolia	Female	PSS	Elderly	Elderly	0	0	PS
3	Sonia	Female	PSS	26	Adult	0	0	
4	Marnie	Female	PSS	Teenager	Teenager	0	0	

	dtype	missing_count	missing_rate
Series	object	323	50.710
Age_numeric	float64	136	21.350
Age	object	136	21.350
Age_range	object	136	21.350
Destructoid	float64	64	10.050
IGN	float64	32	5.020
GameSpot	float64	29	4.550
Species	object	5	0.780
Release_Date	datetime64[ns]	0	0.000
Metacritic	float64	0	0.000
Avg_Reviews	float64	0	0.000
Percentage_non_male_num	float64	0	0.000
Team_percentage_num	float64	0	0.000
Release_Year	int32	0	0.000
Release_Month	int32	0	0.000

EDA Summary

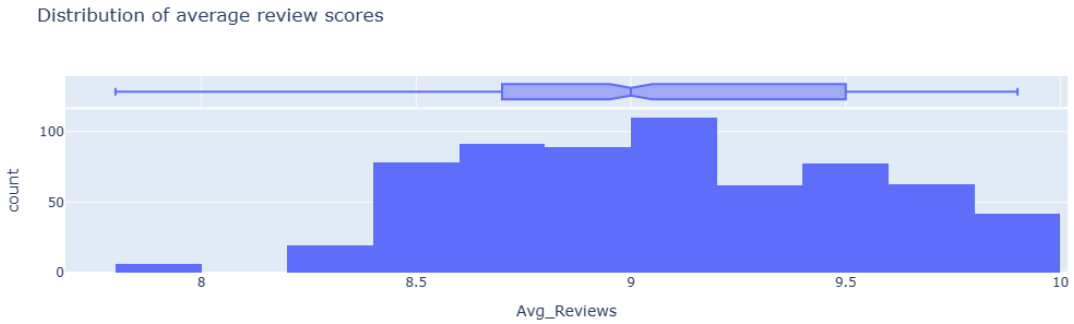
The app's EDA page focuses on score behavior, gender-representation variables, missing data, and simple relationships between numeric features. The next cell

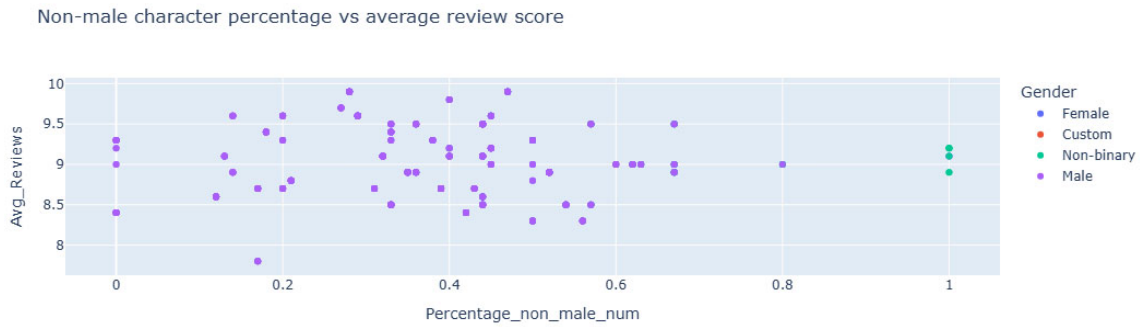
reproduces that style with a few compact plots:

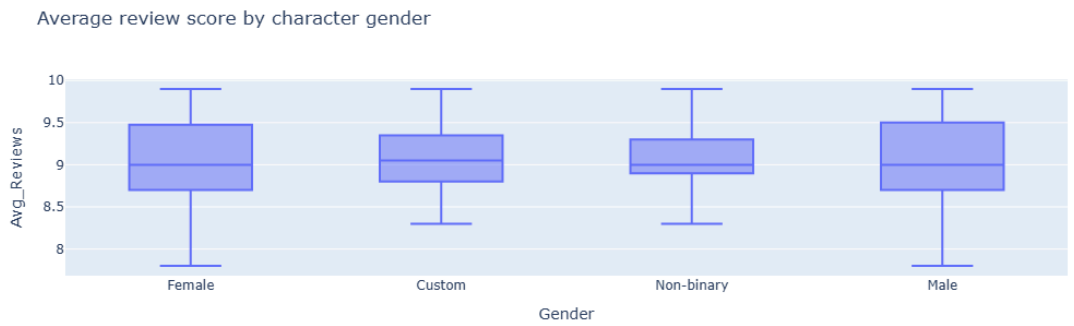
- the distribution of `Avg_Reviews`
- a scatter plot for non-male character share vs average review score
- a box plot of average review score by character gender
- a correlation heatmap for the main modeling features

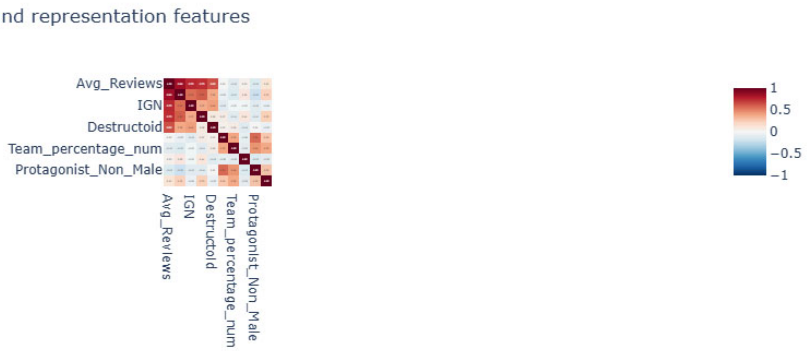
In [4]:
```
eda_features = [
    "Avg_Reviews",
    "Metacritic",
    "IGN",
    "GameSpot",
    "Destructoid",
    "Percentage_non_male_num",
    "Team_percentage_num",
    "Sexualization",
    "Protagonist_Non_Male",
    "Release_Year",
]
eda_features = [column for column in eda_features if column in df.columns]

fig_reviews = px.histogram(
    df,
    x="Avg_Reviews",
    nbins=20,
    marginal="box",
    title="Distribution of average review scores",
)
fig_reviews.show()

if {"Percentage_non_male_num", "Avg_Reviews"}.issubset(df.columns):
    fig_representation = px.scatter(
        df,
        x="Percentage_non_male_num",
        y="Avg_Reviews",
        color="Gender" if "Gender" in df.columns else None,
        hover_name="Title" if "Title" in df.columns else None,
        title="Non-male character percentage vs average review score",
    )
    fig_representation.show()

if {"Gender", "Avg_Reviews"}.issubset(df.columns):
    fig_gender_box = px.box(
        df,
        x="Gender",
        y="Avg_Reviews",
        points="outliers",
        title="Average review score by character gender",
    )
    fig_gender_box.show()

correlation_df = df[eda_features].corr(numeric_only=True)
fig_heatmap = px.imshow(
    correlation_df,
    text_auto=".2f",
    color_continuous_scale="RdBu_r",
    zmin=-1,
    zmax=1,
```

```
    title="Correlation heatmap for score and representation features",
)
fig_heatmap.show()

display(df[eda_features].describe().T)
```









	count	mean	std	min	25%	50%
Avg_Reviews	637.000	9.051	0.451	7.800	8.700	9.000
Metacritic	637.000	8.917	0.474	7.900	8.700	8.900
IGN	605.000	9.297	0.544	8.000	9.000	9.300
GameSpot	608.000	8.855	0.782	7.000	8.500	9.000
Destructoid	573.000	9.125	0.687	7.000	8.600	9.000
Percentage_non_male_num	637.000	0.383	0.175	0.000	0.280	0.390
Team_percentage_num	637.000	0.109	0.178	0.000	0.000	0.000
Sexualization	637.000	0.086	0.389	0.000	0.000	0.000
Protagonist_Non_Male	637.000	0.501	0.554	0.000	0.000	0.000
Release_Year	637.000	2016.911	3.081	2012.000	2014.000	2017.000

Metrics Guide

For the regression models:

- **RMSE**: penalizes larger errors more strongly, so it is useful when we care about big misses.
- **MAE**: measures the average absolute miss and is easier to interpret in score units.
- **R2**: estimates how much of the variation in review scores is explained by the model.

For the Naive Bayes classifier:

- **Accuracy**: overall share of correct predictions.
- **Precision**: among games predicted as high-scoring, how many truly are high-scoring.
- **Recall**: among truly high-scoring games, how many the classifier catches.
- **F1**: a balance between precision and recall.

Naive Bayes is a classifier rather than a regressor, so we turn `Avg_Reviews` into a binary label: high score (`>= 9.0`) vs lower score (`< 9.0`). The `9.0` threshold is used here because a lower cutoff such as `7.5` would collapse this dataset into almost a single class.

In [5]:
```
target = "Avg_Reviews"
regression_features = [
    "Metacritic",
    "IGN",
    "GameSpot",
    "Destructoid",
    "Percentage_non_male_num",
    "Sexualization",
```

```
    "Protagonist_Non_Male",
    "Team_percentage_num",
    "Release_Year",
    "PEGI",
]

model_df = df[regression_features + [target]].copy()
X = model_df[regression_features]
y = model_df[target]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=TEST_SIZE,
    random_state=RANDOM_STATE,
)

print(f"Training rows: {len(X_train)}")
print(f"Testing rows: {len(X_test)}")
print(f"Target: {target}")
print(f"Features: {', '.join(regression_features)}")
```

```
Training rows: 445
Testing rows: 192
Target: Avg_Reviews
Features: Metacritic, IGN, GameSpot, Destructoid, Percentage_non_male_num, Sexual
ization, Protagonist_Non_Male, Team_percentage_num, Release_Year, PEGI
```

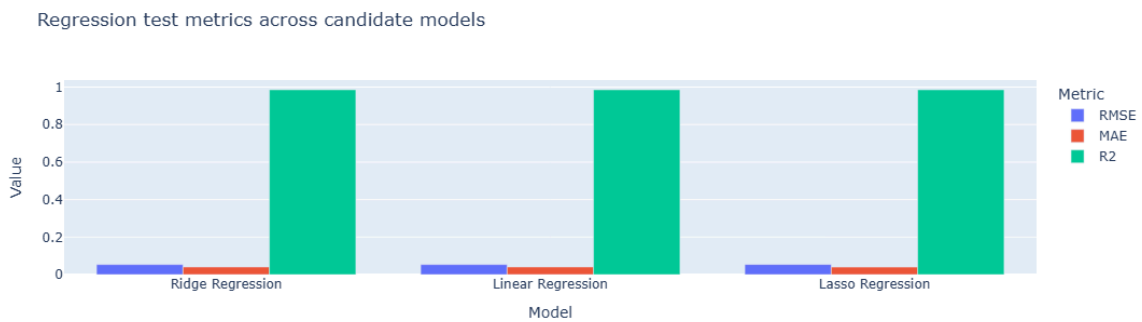
In [6]:

```
regression_models = {
    "Linear Regression": LinearRegression(),
    "Ridge Regression": RidgeCV(alphas=np.logspace(-3, 2, 60)),
    "Lasso Regression": LassoCV(
        alphas=np.logspace(-3, 1, 60),
        cv=5,
        max_iter=20000,
        random_state=RANDOM_STATE,
    ),
}

fitted_regression_models = {}
regression_rows = []

for model_name, estimator in regression_models.items():
    pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler()),
        ("model", estimator),
    ])
    pipeline.fit(X_train, y_train)
    predictions = pipeline.predict(X_test)

    row = {
        "Model": model_name,
        "RMSE": mean_squared_error(y_test, predictions) ** 0.5,
        "MAE": mean_absolute_error(y_test, predictions),
        "R2": r2_score(y_test, predictions),
    }
    if hasattr(pipeline.named_steps["model"], "alpha_"):
        row["Selected alpha"] = pipeline.named_steps["model"].alpha_
```

```
    regression_rows.append(row)
    fitted_regression_models[model_name] = pipeline

regression_results = pd.DataFrame(regression_rows).sort_values("RMSE").reset_ind
display(regression_results)
```

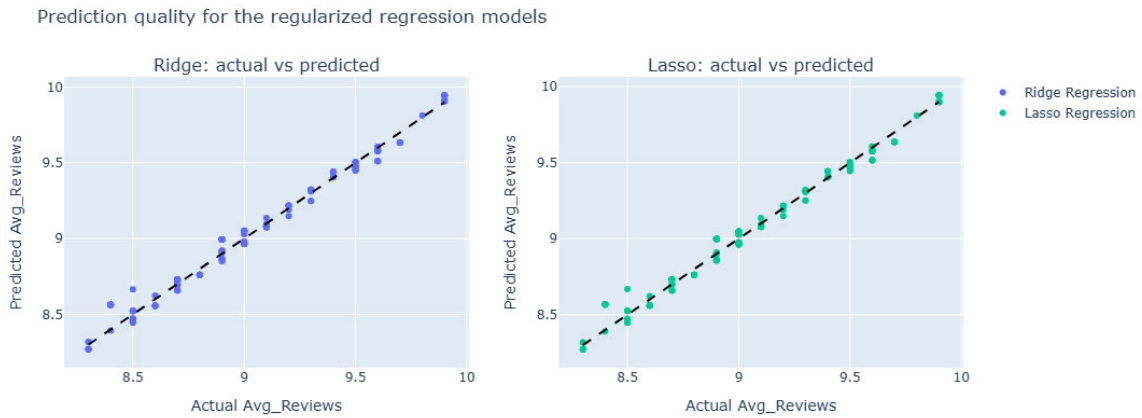
	Model	RMSE	MAE	R2	Selected alpha
0	Ridge Regression	0.053	0.040	0.986	0.424
1	Linear Regression	0.053	0.040	0.986	NaN
2	Lasso Regression	0.054	0.040	0.985	0.001

In [7]:
```
metric_long_df = regression_results.melt(
    id_vars="Model",
    value_vars=["RMSE", "MAE", "R2"],
    var_name="Metric",
    value_name="Value",
)

fig_metric_bars = px.bar(
    metric_long_df,
    x="Model",
    y="Value",
    color="Metric",
    barmode="group",
    title="Regression test metrics across candidate models",
)
fig_metric_bars.show()
```



In [8]:
```
fig_predictions = make_subplots(
    rows=1,
    cols=2,
    subplot_titles=("Ridge: actual vs predicted", "Lasso: actual vs predicted"),
)

for index, model_name in enumerate(["Ridge Regression", "Lasso Regression"], sta
    predictions = fitted_regression_models[model_name].predict(X_test)
    fig_predictions.add_trace(
        go.Scatter(
            x=y_test,
            y=predictions,
            mode="markers",
            name=model_name,
        ),
        row=1,
```

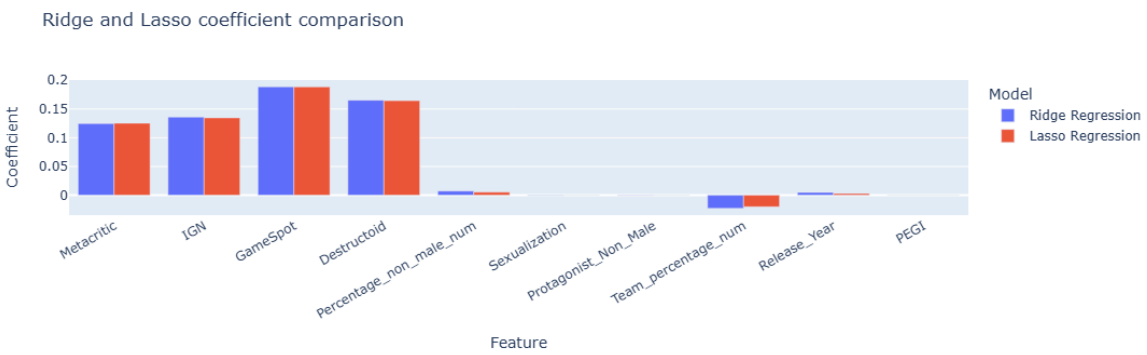
```
        col=index,
    )
    fig_predictions.add_trace(
        go.Scatter(
            x=[y_test.min(), y_test.max()],
            y=[y_test.min(), y_test.max()],
            mode="lines",
            line=dict(color="black", dash="dash"),
            showlegend=False,
        ),
        row=1,
        col=index,
    )

fig_predictions.update_xaxes(title_text="Actual Avg_Reviews")
fig_predictions.update_yaxes(title_text="Predicted Avg_Reviews")
fig_predictions.update_layout(height=450, title="Prediction quality for the regu
fig_predictions.show()
```



In [9]:

```
coefficient_rows = []
for model_name in ["Ridge Regression", "Lasso Regression"]:
    coefficients = fitted_regression_models[model_name].named_steps["model"].coe
    coefficient_rows.extend(
        {
            "Feature": feature,
            "Coefficient": coefficient,
            "Model": model_name,
        }
        for feature, coefficient in zip(regression_features, coefficients)
    )

coefficient_df = pd.DataFrame(coefficient_rows)
display(coefficient_df.sort_values(["Model", "Coefficient"], ascending=[True, Fa

fig_coefficients = px.bar(
    coefficient_df,
    x="Feature",
    y="Coefficient",
    color="Model",
    barmode="group",
    title="Ridge and Lasso coefficient comparison",
)
fig_coefficients.update_layout(xaxis_tickangle=-30)
fig_coefficients.show()
```

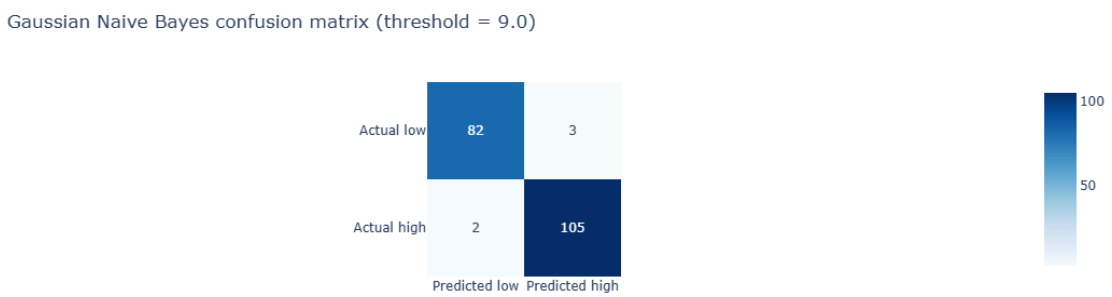
	Feature	Coefficient	Model
12	GameSpot	0.188	Lasso Regression
13	Destructoid	0.164	Lasso Regression
11	IGN	0.135	Lasso Regression
10	Metacritic	0.125	Lasso Regression
14	Percentage_non_male_num	0.005	Lasso Regression
18	Release_Year	0.003	Lasso Regression
15	Sexualization	-0.000	Lasso Regression
16	Protagonist_Non_Male	0.000	Lasso Regression
19	PEGI	0.000	Lasso Regression
17	Team_percentage_num	-0.020	Lasso Regression
2	GameSpot	0.189	Ridge Regression
3	Destructoid	0.165	Ridge Regression
1	IGN	0.136	Ridge Regression
0	Metacritic	0.125	Ridge Regression
4	Percentage_non_male_num	0.007	Ridge Regression
8	Release_Year	0.005	Ridge Regression
9	PEGI	0.000	Ridge Regression
6	Protagonist_Non_Male	-0.000	Ridge Regression
5	Sexualization	-0.000	Ridge Regression
7	Team_percentage_num	-0.023	Ridge Regression



In [10]:
```
nb_target = (df[target] >= NB_HIGH_SCORE_THRESHOLD).astype(int)
X_train_nb, X_test_nb, y_train_nb, y_test_nb = train_test_split(
    X,
    nb_target,
    test_size=TEST_SIZE,
    random_state=RANDOM_STATE,
    stratify=nb_target,
)

naive_bayes_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
```

```
    ("scaler", StandardScaler()),
    ("model", GaussianNB()),
])
naive_bayes_pipeline.fit(X_train_nb, y_train_nb)
nb_predictions = naive_bayes_pipeline.predict(X_test_nb)

nb_results = pd.DataFrame([
    {
        "Metric": "Accuracy",
        "Value": accuracy_score(y_test_nb, nb_predictions),
    },
    {
        "Metric": "Precision",
        "Value": precision_score(y_test_nb, nb_predictions),
    },
    {
        "Metric": "Recall",
        "Value": recall_score(y_test_nb, nb_predictions),
    },
    {
        "Metric": "F1",
        "Value": f1_score(y_test_nb, nb_predictions),
    },
])
display(nb_results)

nb_confusion = confusion_matrix(y_test_nb, nb_predictions)
fig_nb_confusion = px.imshow(
    nb_confusion,
    text_auto=True,
    color_continuous_scale="Blues",
    x=["Predicted low", "Predicted high"],
    y=["Actual low", "Actual high"],
    title=f"Gaussian Naive Bayes confusion matrix (threshold = {NB_HIGH_SCORE_TH
)
fig_nb_confusion.show()
```

	Metric	Value
0	Accuracy	0.974
1	Precision	0.972
2	Recall	0.981
3	F1	0.977



Requested Naive Bayes Analysis Using IGN Thresholds

This section follows the requested rule exactly:

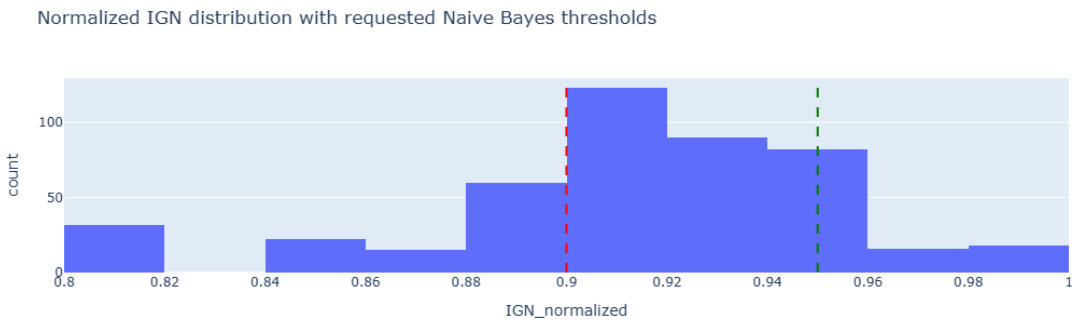
- `high` if normalized `IGN > 0.95`
- `low` if normalized `IGN < 0.9`

To make the thresholds match the `0.0` to `1.0` scale, the notebook divides `IGN` by `10`. Games in the middle band (`0.7` to `0.9`) are excluded from this specific experiment.

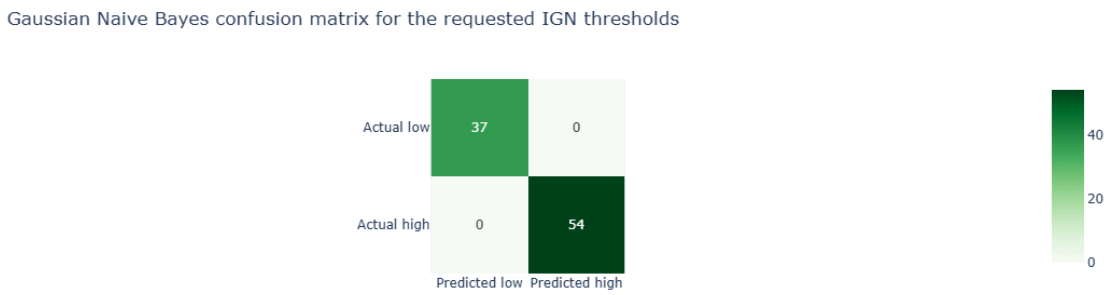
In [11]:
```
ign_nb_columns = list(dict.fromkeys(regression_features + ["IGN"]))
ign_nb_df = df[ign_nb_columns].dropna().copy()
ign_nb_df["IGN_normalized"] = ign_nb_df["IGN"] / 10

ign_nb_df = ign_nb_df.loc[
    (ign_nb_df["IGN_normalized"] > IGN_HIGH_THRESHOLD)
    | (ign_nb_df["IGN_normalized"] < IGN_LOW_THRESHOLD)
].copy()

ign_nb_df["IGN_label"] = np.where(
    ign_nb_df["IGN_normalized"] > IGN_HIGH_THRESHOLD,
    "high",
    "low",
)

class_counts = ign_nb_df["IGN_label"].value_counts().rename_axis("Class").reset_
display(class_counts)

fig_ign_thresholds = px.histogram(
    df.assign(IGN_normalized=df["IGN"] / 10),
    x="IGN_normalized",
    nbins=20,
    title="Normalized IGN distribution with requested Naive Bayes thresholds",
)
fig_ign_thresholds.add_vline(x=IGN_LOW_THRESHOLD, line_dash="dash", line_color="
fig_ign_thresholds.add_vline(x=IGN_HIGH_THRESHOLD, line_dash="dash", line_color=
fig_ign_thresholds.show()

if ign_nb_df["IGN_label"].nunique() < 2:
    print(
        "The requested IGN-threshold split cannot train Gaussian Naive Bayes on
        "because only one class is present after filtering."
    )
else:
    X_ign_nb = ign_nb_df[regression_features]
    y_ign_nb = (ign_nb_df["IGN_label"] == "high").astype(int)

    X_train_ign_nb, X_test_ign_nb, y_train_ign_nb, y_test_ign_nb = train_test_sp
        X_ign_nb,
        y_ign_nb,
        test_size=TEST_SIZE,
        random_state=RANDOM_STATE,
        stratify=y_ign_nb,
    )
```

```
ign_nb_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler()),
    ("model", GaussianNB()),
])
ign_nb_pipeline.fit(X_train_ign_nb, y_train_ign_nb)
ign_nb_predictions = ign_nb_pipeline.predict(X_test_ign_nb)

ign_nb_results = pd.DataFrame([
    {"Metric": "Accuracy", "Value": accuracy_score(y_test_ign_nb, ign_nb_pre
    {"Metric": "Precision", "Value": precision_score(y_test_ign_nb, ign_nb_p
    {"Metric": "Recall", "Value": recall_score(y_test_ign_nb, ign_nb_predict
    {"Metric": "F1", "Value": f1_score(y_test_ign_nb, ign_nb_predictions)},
])
display(ign_nb_results)

ign_nb_confusion = confusion_matrix(y_test_ign_nb, ign_nb_predictions)
fig_ign_nb_confusion = px.imshow(
    ign_nb_confusion,
    text_auto=True,
    color_continuous_scale="Greens",
    x=["Predicted low", "Predicted high"],
    y=["Actual low", "Actual high"],
    title="Gaussian Naive Bayes confusion matrix for the requested IGN thres
)
fig_ign_nb_confusion.show()
```

	Class	Count
0	high	179
1	low	123



	Metric	Value
0	Accuracy	1.000
1	Precision	1.000
2	Recall	1.000
3	F1	1.000



Key Takeaways

- Ridge and Lasso provide the main regression comparison, with the held-out test split judged by RMSE, MAE, and R2.
- The coefficient plot helps explain which features contribute most strongly after scaling and regularization.
- Naive Bayes is included as a separate classification view of the problem: it predicts whether a game falls into the high-score group rather than estimating the exact review score.
- The additional IGN-threshold Naive Bayes analysis now checks the requested rule of `IGN > 0.95` for high and `IGN < 0.9` for low after normalizing IGN to a `0` to `1` scale, while excluding the middle band.
- Because `Avg_Reviews` is closely tied to the critic-site scores (`Metacritic`, `IGN`, `GameSpot`, `Destructoid`), those features will typically dominate predictive performance. Representation-related variables still appear in the workflow, but they should be interpreted as supporting signals rather than the sole driver of the score.